Parseval's Theorem

$\{x(n)\} \xleftrightarrow{F} X(\omega)$

$\{y(n)\} \xleftrightarrow{F} Y(\omega)$

$$\sum_{n=-\infty}^{\infty} x(n)\, y^*(n) = \frac{1}{2\pi} \int_{\omega=-\pi}^{\pi} X(\omega)\, Y^*(\omega)\, d\omega$$

inner product or dot product

inner product.

Proof.

$$y(n) = \frac{1}{2\pi} \int_{\omega=-\pi}^{\pi} Y(\omega)\, e^{j\omega n}\, d\omega$$

$$y^*(n) = \frac{1}{2\pi} \int_{\omega=-\pi}^{\pi} Y^*(\omega)\, e^{-j\omega n}\, d\omega$$

$$\sum_n x(n)\, y^*(n) = \frac{1}{2\pi} \int_{\omega=-\pi}^{\pi} Y^*(\omega) \underbrace{\sum_n x(n) e^{-j\omega n}}_{X(\omega)}\, d\omega$$

$$= \frac{1}{2\pi} \int_{\omega=-\pi}^{\pi} X(\omega)\, Y^*(\omega)\, d\omega$$

Special case $Y(\omega) = X(\omega)$, $y(n) = x(n)$, $y^*(n) = x^*(n)$

$$\sum_n x(n)\, x^*(n) = \frac{1}{2\pi} \int_{\omega=-\pi}^{\pi} X(\omega)\, X^*(\omega)\, d\omega$$

$$\boxed{\sum_n |x(n)|^2 = \frac{1}{2\pi} \int_{\omega=-\pi}^{\pi} |X(\omega)|^2\, d\omega.}$$

Filtering (Application of convolution property)

LTI System
impulse response $\{h(n)\}$

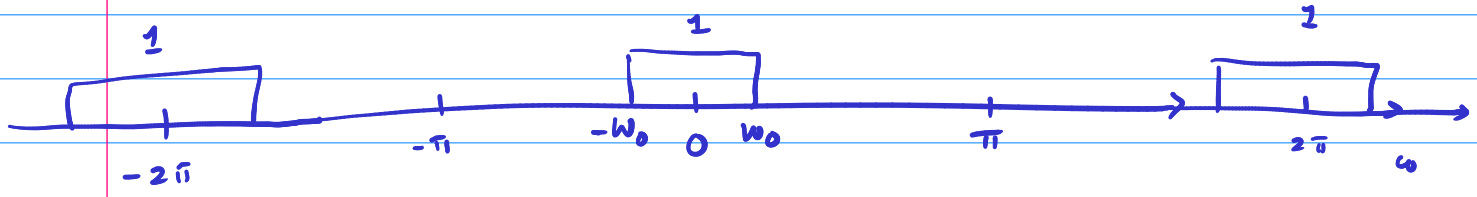
$$\{y(n)\} = \{x(n)\} * \{h(n)\}$$

$$Y(\omega) = X(\omega)\, \underset{\sim}{H(\omega)}$$

Frequency response of System

$H(\omega)$ → gain given by LTI System for frequency ω.

(1) Ideal Low Pass filter



$$H(\omega)_{LP} = \begin{cases} 1 ; & |\omega| \le \omega_0 \\ 0 ; & \omega_0 < |\omega| \le \pi \end{cases}$$

$$h_{LP}(n) \xleftrightarrow{\mathcal{F}} H_{LP}(\omega)$$

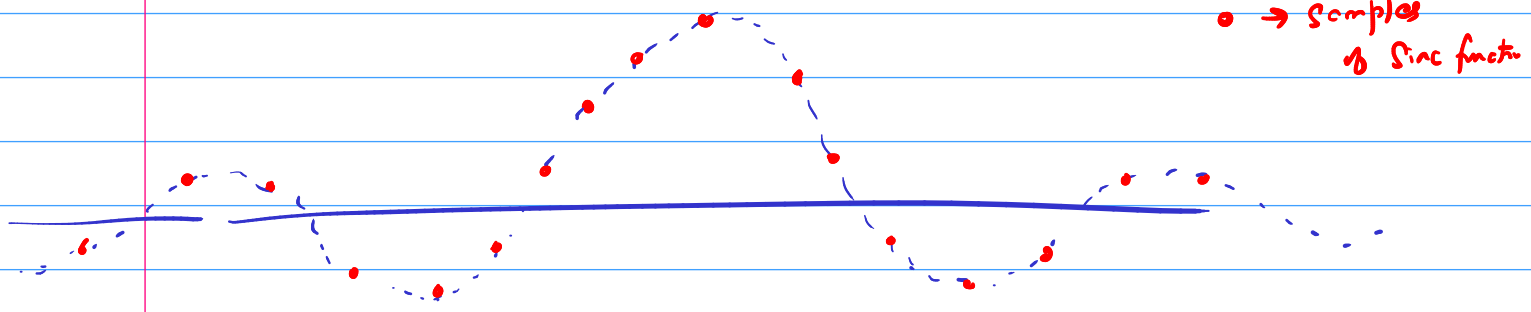
impulse response of ideal low pass filter (LPF)

$$h_{LP}(n) = \frac{1}{2\pi} \int_{\omega=-\omega_0}^{\omega_0} 1 \cdot e^{j\omega n}\, d\omega$$

$$= \frac{1}{2\pi} \left. \frac{e^{j\omega n}}{jn} \right|_{-\omega_0}^{\omega_0}$$

$$= \frac{1}{\pi n} \frac{1}{2j} \underbrace{\left(e^{j\omega_0 n} - e^{-j\omega_0 n} \right)}$$

$$\boxed{h_{LP}(n) = \frac{1}{\pi n} \sin(\omega_0 n)}$$



$h_{LP}(n)$ is not absolutely summable

$$\sum_n \frac{1}{n} = \infty$$

It is unstable system

$$\sum_n \frac{1}{n^2} < \infty$$

Energy

$$\sum_n |h_{LP}(n)|^2 = \frac{1}{2\pi} \int_{-\omega_0}^{\omega_0} 1^2 \, d\omega$$

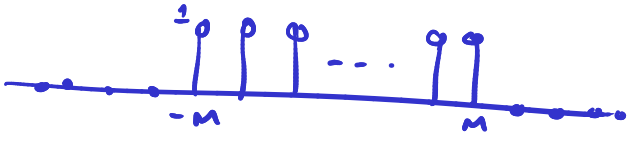
Parseval's Thm

$$= \frac{2\omega_0}{2\pi} = \frac{\omega_0}{\pi}$$

2. Window signal

Rectangular Window $w(n) = \begin{cases} 1 & ; |n| \le M \\ 0 & ; |n| > M \end{cases}$

(Impulse response of Moving Average filter)



Spectrum of window

$$W(\omega) = \sum_{n=-M}^{M} e^{-j\omega n} \qquad \alpha = e^{-j\omega}$$

$\alpha \in \mathbb{C}$

$$\sum_{n=-N_1}^{N_2} \alpha^n = \frac{\alpha^{-N_1} - \alpha^{N_2+1}}{1-\alpha}$$

$N_2 \geq N_1$

$$= \frac{e^{+j\omega M} - e^{-j\omega(M+1)}}{1 - e^{-j\omega}}$$

$$= \frac{e^{-j\omega/2} e^{+j\omega/2} e^{+j\omega M} - e^{-j\omega M} e^{-j\omega/2} e^{-j\omega/2}}{e^{-j\frac{\omega}{2}} e^{+j\omega/2} - e^{-j\omega/2} e^{-j\omega/2}}$$

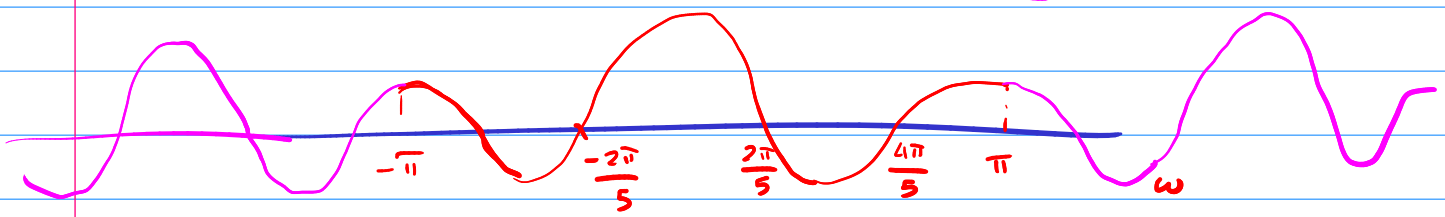
$$= \frac{e^{j\frac{\omega}{2}(2M+1)} - e^{-j\frac{\omega}{2}(2M+1)}}{e^{j\omega/2} - e^{-j\omega/2}}$$

$$W(\omega) = \frac{\sin\left(\frac{\omega}{2}(2M+1)\right)}{\sin(\omega/2)}$$

Dirichlet kernel

$M = 2$ $\quad W(\omega) = \dfrac{\sin\left(\frac{5\omega}{2}\right)}{\sin(\omega/2)}$

Periodic Sinc signal.



(3) Truncated Low pass filter

$$h_{TLP}(n) = \begin{cases} h_{LP}(n) & ; \ |n| \le M \\ 0 & ; \ \text{else} \end{cases}$$

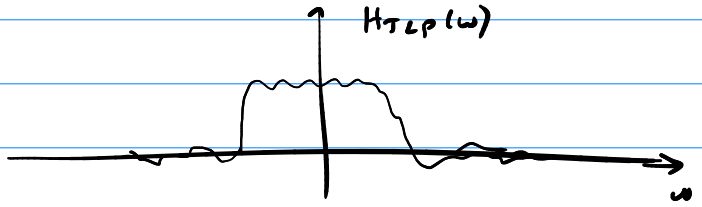
$$= h_{LP}(n) \cdot w(n)$$

$$H_{TLP}(\omega) = H_{LP}(\omega) * W(\omega)$$

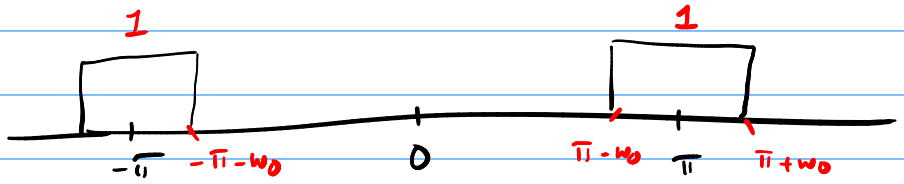
Periodic convolution

$$= \frac{1}{2\pi} \int_{\theta=-\pi}^{\pi} H_{LP}(\theta)\, W(\omega-\theta)\, d\theta$$

When M is large



(4) Ideal High Pass filter



$$H_{HP}(\omega) = H_{LP}(\omega - \pi)$$

$$h_{HP}(n) = h_{LP}(n)\, e^{j\pi n}$$

$$\boxed{h_{HP}(n) = (-1)^n h_{LP}(n)}$$